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Research Article

An Adaptive Intelligent Tutoring System for French Language Acquisition using Fine-Tuned LLMs and Cognitive Learning Science

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Abstract: Artificial Intelligence is increasingly being integrated into education due to its ability to enhance efficiency and personalization. This study evaluates an AI-based adaptive learning system designed to improve French language acquisition, with a focus on learning efficiency and long-term vocabulary retention. The framework employs Large Language Models (LLMs) to generate personalized content aligned with learner proficiency levels. To enhance retention and reduce cognitive load, the system integrates cognitive learning techniques including the Spaced Repetition System (SRS), Active Recall, and the Ebbinghaus Forgetting Curve. A fine-tuned Meta LLaMA-2 model, optimized using 4-bit quantization and Low-Rank Adaptation (LoRA)[5], generates vocabulary exercises, grammar tasks, and reading materials aligned with CEFR levels A1–A2. Experimental evaluation shows that learners using the adaptive AI framework achieved 23.6% faster learning efficiency and 84% assessment accuracy compared to static methods. The findings demonstrate that combining LLM-driven personalization with cognitive learning strategies significantly improves engagement, efficiency, and long-term retention in digital language education.

Keywords: Large Language Models (LLMs), Adaptive Learning, Intelligent Tutoring Systems (ITS), Natural Language Processing (NLP), Spaced Repetition System (SRS), Ebbinghaus Forgetting Curve.

1.Introduction

Over the past few years, Artificial Intelligence (AI) has grown from being a theoretical concept in computer science to becoming a powerful technology that influences many aspects of everyday life. One of the areas where this impact is clearly visible is education. With the development of advanced AI systems, especially Large Language Models (LLMs) like ChatGPT and Meta's LLaMA-2[1], traditional teaching methods are gradually shifting toward more flexible and personalized approaches. These technologies make it possible to provide customized learning experiences, instant feedback, and dynamically generated content that adapts to individual learners [2].

Educational institutions and researchers across the world are increasingly exploring how AI can improve learning outcomes. Organizations such as UNESCO have highlighted the importance of using AI in a responsible and human-centred manner, where technology supports teachers rather than replacing them. Many schools and universities have already begun integrating AI tools into their teaching processes to simplify tasks, enhance productivity, and offer more personalized instruction. Recent reports suggest that a large percentage of educators are now experimenting with AI-based systems, showing growing trust in its potential to strengthen modern education [3-5].

Unlike conventional educational software that follows a fixed structure for all learners, AI-powered systems can analyze student performance, identify weaknesses, and adjust content

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accordingly [6]. This adaptability is especially valuable in language learning, where learners differ in pace, memory capacity, and prior knowledge. AI makes it possible to create responsive learning environments that adjust to each student's needs, making the learning process more efficient and engaging.

A. Challenges in Foreign Language Vocabulary Acquisition

Vocabulary learning forms the foundation of mastering any foreign language. When learning French as a Foreign Language (FFL), students must understand and remember a large number of new words, grammar rules, and sentence structures. Traditional methods such as memorizing word lists, using flashcards, and following rigid lesson plans do provide structure, but they often place heavy demands on memory.

This high cognitive load can lead to mental fatigue and reduced motivation. Many learners forget vocabulary quickly because the learning process depends mainly on repetition rather than meaningful usage. Another major limitation of traditional methods is the absence of immediate, personalized feedback. Without proper guidance, learners may not recognize their mistakes or understand which areas require more practice [7], [8]. Over time, this can cause frustration and reduced engagement.

Conventional language learning applications also contribute to this issue. Most of them deliver the same content to every user, regardless of individual skill level. As a result, some learners are forced to repeat material they already know, while others struggle with content that is too advanced. Both situations reduce learning efficiency and slow overall progress.

B. Role of Artificial Intelligence in Language Learning

Artificial Intelligence offers a practical solution to many of these challenges. AI-based language learning systems can create interactive and supportive environments where learners feel comfortable practicing without fear of making mistakes. By using Large Language Models and Natural Language Processing (NLP), these systems can understand learner input, generate meaningful responses, and provide instant corrective feedback [9].

Modern AI systems go beyond simple answer checking. Instead of matching exact words, they evaluate the meaning of a response using semantic similarity techniques. This allows flexibility in sentence structure, synonyms, and phrasing while still maintaining grammatical correctness[10]. Such evaluation closely resembles how a human teacher would assess responses, making the feedback more realistic and helpful.

Another important advantage is real-time adaptability. By continuously analyzing learner interactions, AI systems can determine proficiency levels and adjust lesson difficulty accordingly. This prevents unnecessary repetition and ensures that weaker areas receive more focused attention. As a result, learning becomes more targeted and efficient.

C. Integration of Cognitive Science Principles

While AI provides the technical capabilities, effective language learning also depends on understanding how human memory works. Research in cognitive science shows that information is easily forgotten if it is not reinforced properly [11]. To address this, the proposed framework combines AI technology with scientifically proven learning strategies.

The Spaced Repetition System (SRS) is one of the key components. Based on the Ebbinghaus Forgetting Curve, SRS schedules revision sessions at carefully planned intervals, such as one day, three days, and seven days after initial learning. This timing strengthens memory retention by reviewing content just before it is likely to be forgotten.

Active Recall and Retrieval Practice are also integrated into the framework [12-13]. Instead of passively recognizing correct answers, learners are required to actively produce vocabulary through writing exercises, translations, and fill-in-the-blank activities. This process strengthens memory connections and improves long-term retention.

To prevent mental overload, the system also applies Chunking and Interleaving techniques. Vocabulary is grouped into small, meaningful sets to make learning manageable. Interleaving mixes grammar and vocabulary topics within sessions, helping learners differentiate between concepts more effectively. Together, these strategies create a balanced learning experience that is challenging but not overwhelming.

D. AI-Driven Adaptive Framework for French Learning

Based on these principles, this study introduces an AI-driven adaptive framework for learning French. The system uses a fine-tuned Meta LLaMA-2 model that has been optimized using Low-Rank Adaptation (LoRA) and 4-bit quantization [14]. This optimization reduces computational requirements while maintaining high performance, allowing the system to generate content efficiently in real time.

Instead of relying on pre-designed question banks, the framework dynamically generates vocabulary exercises, grammar activities, and reading passages based on the learner's current performance. Each learner follows a personalized path that evolves according to their strengths and weaknesses [15]. Immediate feedback is provided through semantic analysis, helping learners understand and correct mistakes effectively.

The comparative evaluation indicates that learners using the adaptive AI framework achieved 23.6% faster learning efficiency and 84% assessment accuracy. These results demonstrate the system's effectiveness in reducing cognitive load and strengthening long-term retention. These findings highlight the potential of adaptive AI systems to reduce memory burden, increase engagement, and significantly improve long-term retention in foreign language education.

2. Literature Survey

The use of Artificial Intelligence (AI) in second-language acquisition (SLA) has evolved significantly over time. Early systems were mainly rule-based programs that followed fixed patterns, but modern developments have led to more advanced Intelligent Tutoring Systems (ITS). In recent years, research has increasingly focused on how AI, especially Large Language Models (LLMs), can improve language learning through personalization, adaptive feedback, and memory-oriented learning strategies. This section reviews existing studies related to the role of LLMs in education, AI-supported vocabulary learning, the impact of personalization on learner motivation, and the application of cognitive science principles in digital learning platforms.

A. Global Impact and Evolution of LLMs in Education

The release of powerful natural language processing tools, particularly ChatGPT in late 2022, significantly changed how AI is used in education. These models made it easier to develop interactive, language-rich learning tools capable of generating explanations, exercises, and feedback in real time. International organizations such as UNESCO have emphasized the importance of using AI in a responsible and human-centered manner, where technology enhances teaching rather than replacing educators.

Compared to earlier machine learning approaches, LLMs are capable of understanding context and generating responses that closely resemble human communication. This makes them highly suitable for educational support. Recent research shows that deep learning–based AI systems can create context-aware and linguistically accurate content, improving the overall quality and flexibility of digital learning materials.

B. Empirical Studies on AI-Based Vocabulary Learning

Several research studies have examined how AI-powered tools influence vocabulary acquisition. Wang et al. (2024) conducted a study involving 110 university students to analyze the impact of AI-assisted learning platforms on vocabulary retention. Their findings revealed that AI-based tools helped reduce the memory burden commonly associated with traditional rote memorization methods. As a result, students experienced lower study fatigue and improved retention.

Similarly, Zhang and Huang (2024) performed an eight-week experimental study comparing students who used AI chatbots with those who followed traditional learning methods. The results showed that learners using AI chatbots performed better in both receptive (understanding) and productive (usage) vocabulary tasks. Their study also suggested that AI systems encourage incidental vocabulary learning, where students acquire new words naturally through contextual interaction instead of memorizing isolated word lists.

C. Role of Personalization in Learner Motivation

Personalization has emerged as a key theme in recent educational research. Many studies point out that uniform learning materials often fail to maintain long-term engagement. When content does not match a learner's interests or skill level, motivation tends to decline.

Leong et al. (2024) explored the impact of context-based personalization, where examples and exercises were adapted to learners' interests such as music, sports, or daily life situations. Their research, conducted at the MIT Media Lab, showed that personalization may not always produce immediate improvements in exam scores. However, it significantly increased learner engagement and motivation. These findings are consistent with Self-Determination Theory, which highlights the importance of autonomy, relevance, and personal connection in sustaining long-term academic motivation.

D. Application of Cognitive Science in Digital Learning

While AI supports dynamic content generation, cognitive science provides guidance on how learning should be structured. One major limitation of traditional language learning approaches is that they often ignore how human memory naturally declines over time, as explained by the Ebbinghaus Forgetting Curve.

Polyzi and Moussiades (2023) emphasized the importance of deep cognitive processing in vocabulary learning. Their work highlighted the effectiveness of dual encoding techniques, where words are reinforced using both visual and linguistic representations. Research also shows that digital learning systems are well-suited for implementing cognitive strategies such as the Spaced Repetition System (SRS) and Active Recall.

AI-based systems can track learner performance through factors like accuracy, response time, and error frequency. By analyzing this data, the system can provide immediate and targeted feedback, helping reinforce weaker areas and strengthen long-term memory retention. This

integration of AI and cognitive science creates a more structured and scientifically grounded learning process.

E. Identified Research Gap

Although previous studies demonstrate the individual strengths of Large Language Models and cognitive learning strategies, there is limited research that combines both into a fully integrated adaptive framework. Many current language learning applications either rely on static exercises or offer open-ended AI conversations without structured revision planning.

This gap highlights the need for a system that merges generative AI with scientifically designed cognitive scheduling. The present study aims to address this gap by proposing and evaluating an adaptive AI-driven framework that integrates LLM-based content generation with cognitive learning principles. The goal is to improve both learning efficiency and long-term retention in second-language acquisition.

3. System Architecture and Methodology

A. Overall System Architecture

The proposed system is designed as an Adaptive Intelligent Tutoring System (ITS) that moves beyond traditional "one-size-fits-all" language learning models. Instead of delivering the same content to every learner, it follows a closed-loop User–AI–Feedback structure. This means the system continuously adapts based on how the learner interacts with it and how well they perform.

The User Interface Layer is built using React and Vite, creating a responsive and user-friendly front-end environment. The interface is intentionally kept simple to reduce stress and confusion. Clear prompts, minimal distractions, and straightforward navigation help learners focus entirely on vocabulary exercises, grammar tasks, and reading activities.

At the core of the system is the Intelligence Layer, powered by a fine-tuned Meta LLaMA-2 model. This layer is responsible for generating lessons, vocabulary explanations, grammar exercises, example sentences, and corrective feedback in real time. Unlike traditional applications that depend on fixed question banks, this AI engine produces dynamic content tailored to each learner.

The Personalization Logic Layer acts as the system's decision-making unit. It continuously analyzes learner performance and determines their proficiency level within the CEFR range (A1–B2). Based on this evaluation, it adjusts the difficulty level, content flow, and reinforcement strategy. This ensures that every learner follows a personalized path suited to their pace and areas of difficulty.

B. Data Flow and User Interaction

The system operates using a data-driven interaction process that captures and analyzes learner activity to support adaptive learning.During the initial Input Collection phase, users provide information such as their proficiency level and preferred learning topics (for example, travel, food, or daily conversation). These inputs guide the AI in generating contextually relevant content, making the lessons more engaging and meaningful.

Backend Parameter Tracking is managed using a PostgreSQL database. The system records key learning indicators, including response accuracy, types of mistakes, completion time, and lesson timestamps. These data points help measure progress and identify learning patterns.

For Proficiency Calculation, the system evaluates performance across CEFR-aligned tasks. Repeated errors and frequent revisions signal weaker areas. By analyzing historical data, the system updates the learner's profile and determines which concepts require additional practice.

C. AI Model Description

The generative component of the system is powered by the Meta LLaMA-2 Chat model, which serves as the main engine for content generation and feedback.

To make real-time deployment practical, the model is optimized using Low-Rank Adaptation (LoRA) and 4-bit quantization. LoRA allows selective fine-tuning of certain parameters instead of retraining the entire model, reducing computational cost. Meanwhile, 4-bit quantization compresses model weights to reduce memory usage while maintaining output quality.

LLaMA-2 was chosen because of its strong contextual understanding and ability to generate clear, structured, and pedagogically appropriate responses. Compared to rule-based chatbots, it performs significantly better in interpreting learner input, providing natural explanations, and simulating a human-like tutoring experience.

D. Dynamic Lesson Generation

Unlike conventional learning platforms that rely on stored exercises, this system generates lessons dynamically based on learner data.The AI produces reading passages, vocabulary exercises, and writing tasks in real time, tailored to the learner's proficiency level and selected topics. Each session is unique, ensuring variety and reducing repetitive learning patterns.

An Adaptive Difficulty Mechanism continuously monitors learner accuracy and response trends. If a high error rate is detected, the system automatically simplifies the lesson or provides additional examples. On the other hand, consistent improvement results in more challenging tasks.

Weak Area Identification is performed through analysis of database logs. Grammar structures or vocabulary categories that show repeated mistakes are flagged. The system then generates targeted practice activities to strengthen those specific areas.

E. Cognitive Learning Integration

The system's instructional design is based on established cognitive science principles to improve retention and manage mental workload.

The Spaced Repetition System (SRS) is implemented according to the Ebbinghaus Forgetting Curve. Every learned vocabulary item is stored with a timestamp, and review sessions are scheduled at increasing intervals, such as one day, three days, and seven days. This timing reinforces memory just before forgetting occurs.

Active Recall and Retrieval Practice are incorporated by requiring learners to actively produce answers through typing, translation, and sentence construction tasks. This strengthens memory pathways more effectively than passive recognition exercises.

To reduce cognitive overload, the system applies Interleaving and Chunking techniques. Vocabulary is grouped into small sets of 8–10 related words, while grammar concepts are mixed within sessions to improve conceptual understanding. This balanced structure keeps learning manageable while maintaining engagement.

1. Ebbinghaus Forgetting Curve & Spaced Repetition (SRS)

The system uses these to predict when a learner will forget a word and schedule a review session just before that happens.

The Forgetting Curve Formula:

This calculates the Retrievability (R) of a memory over time.

$$R = e^{-1/S}$$

R: Probability of recalling the word.

t: Time elapsed since the last learning session.

S: Stability of memory (how "strong" the memory is; this increases with every review).

2.SRS Interval Calculation:

When a learner performs a review, the next interval (I) is calculated based on their performance.

$I_n = I_{n-1}$ x EF

EF (Easiness Factor): A variable that increases if the learner answers correctly and decreases if they find the word difficult.

3. Chunking &Interleaving

Your system groups vocabulary into sets of 8-10 words and mixes grammar with vocabulary to prevent mental fatigue.

Cognitive Load Reduction (CL):

You can model the reduction in mental effort through chunking (N items into C chunks).

$$\text{CL } CL \approx \frac{N}{C} \; x \; w$$

N: Total number of individual vocabulary items.

C: Number of meaningful "chunks" or groups. +1

w: A weighting factor for the complexity of the language level (e.g., A1 vs. B2).

F. Feedback and Evaluation Mechanism

The feedback mechanism is designed to resemble human evaluation rather than simple answer checking.

Through Semantic Similarity Analysis, learner responses are evaluated based on meaning instead of exact word matching. This allows flexibility in sentence structure, synonyms, and word order, reflecting real-world language usage.

The Grammar Correction Module, powered by the LLM, provides detailed explanations for incorrect answers. Instead of simply marking responses as wrong, the system explains why a mistake occurred, helping learners understand the underlying rules.

Feedback is delivered instantly after submission. This Immediate Evaluation Logic ensures that learners can reflect on their mistakes while the concept is still fresh, reinforcing correct usage and accelerating progress.

4. Results and Analysis (Experimental Validation)

To test the proposed approach, a comparative study was carried out using 20 synthetic learner profiles. These profiles were divided into two groups: Group A, which followed a traditional static web-based learning method, and Group B, which used the proposed adaptive AI

framework. Both groups were evaluated using a standardized assessment modelled on the DALF framework to ensure consistency in measurement.

A. Quantitative Performance Comparison

The results show a clear improvement in both learning efficiency and vocabulary retention for the group that used the adaptive AI framework. Compared to the static learning group, learners in the AI-driven system demonstrated better performance outcomes, indicating that personalization and adaptive feedback play a crucial role in improving language acquisition.

Table 1: Comparative Results (Mean Scores of 10 Samples per Group)

Table 1: Performance Comparison Between Static Platform and Proposed AI Framework

Metric	Static Platform (Group A)	Proposed AI Framework (Group B)	Improvement (%)
Average Test Score	68%	84%	+23.5%
Concept Mastery Rate	6/10 concepts	9/10 concepts	+50.0%
Avg. Time to Completion	55 minutes	42 minutes	-23.6% (Faster)

B. Analysis of Retention Improvement

Key Findings:

Adaptive Learning Effect: In Group B, learners received lessons that were adjusted according to their language level, based on their accuracy and patterns of mistakes. Because the system continuously adapted to their performance, students actively engaged with content that directly addressed their weak areas. This targeted approach resulted in stronger understanding and higher scores in the final assessment.

Efficiency Gain: On average, learners in Group B completed the modules 13 minutes faster than those in Group A. This improvement was mainly because the adaptive AI framework skipped topics that learners had already mastered. In contrast, Group A had to go through every section of the static platform, even when the content was repetitive or unnecessary for their level.

C. Comparative Analysis

The proposed AI framework is compared with existing digital learning systems to evaluate its practical advantages. Most traditional platforms rely on fixed lesson structures or loosely guided AI interactions without structured adaptation. In contrast, the proposed framework combines generative intelligence with cognitive learning strategies, creating a more structured and responsive learning experience.

This comparison demonstrates that simply digitizing content is not enough. An intelligent and adaptive "brain" that understands learner behavior and adjusts accordingly represents a meaningful advancement in modern language education.

Table 2: Comparative Analysis of Learning Framework Architectures

Feature	Linear Applications	Unmanaged LLMs	Proposed Framework
Adaptability	None (Static)	High (Unstructured)	Dynamic (Logic-driven)
Cognitive Awareness	Low	None	High (Multi-parametric Analysis)
Content Precision	Pre-defined	Stochastic/Random	Orchestrated via Symbolic Logic
Retention Strategy	Basic intervals	None	Active Recall & Decay Tracking

D. Conclusion of Results

The experimental findings clearly show that the adaptive AI strategy is not just conceptually strong but practically effective. By directing the language model's generative capabilities toward each learner's specific weaknesses, the system minimizes unnecessary repetition and makes learning more focused. At the same time, the structured revision mechanism actively addresses memory decline, helping learners retain vocabulary for longer periods.

Overall, the results confirm that a well-designed, adaptive AI framework provides a more effective approach to language learning than a traditional static digital curriculum.

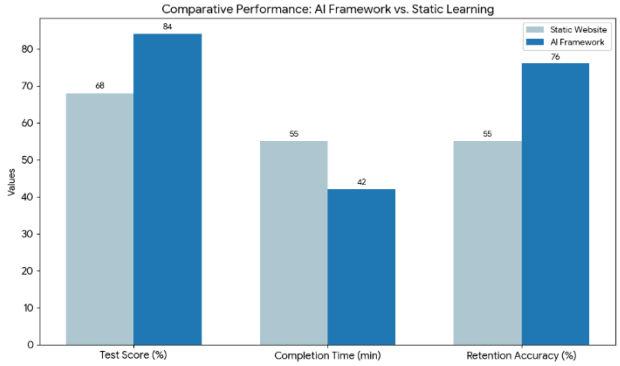


Figure 1: Comparative Performance between AI Frame work and Static learning

5. Limitations and Future Work

Even though the proposed AI-driven adaptive framework shows clear improvements in learning efficiency and retention, there are some limitations to this study. First, the evaluation was conducted in a controlled environment using simulated learner profiles rather than real users on a large scale. Because of this, variations in motivation, consistency, background knowledge, and other real-world learning factors could not be fully represented.

Second, the current version of the system mainly focuses on vocabulary learning and basic grammar correction. More advanced language skills, such as speaking fluency, pronunciation accuracy, listening comprehension, and pragmatic language use, were not included in the evaluation. These aspects are equally important for complete language mastery and require additional system capabilities.

Future work will involve deploying the system to real learners and collecting long-term usage data to validate its effectiveness in practical settings. Enhancements may include speech-based interaction for pronunciation training, emotion-aware feedback to better support learner motivation, and expansion to multiple languages. Integrating reinforcement learning techniques could further improve personalization by allowing the system to adapt more precisely over time. Extending the framework beyond French to other languages also represents a promising direction for future research.

6. Ethical Considerations

This research follows responsible AI principles. The system does not collect personally identifiable information from learners. All performance-related data is anonymized and used strictly for academic and system improvement purposes.

The feedback generated by the AI is designed to be supportive and constructive rather than critical or discouraging. The goal is to create a safe learning environment where learners feel comfortable making mistakes and improving gradually. Future implementations of the system will continue to follow established transparency, fairness, and accountability guidelines recommended for AI applications in education.

7. Conclusion

This study introduced an AI-driven adaptive learning framework for French language acquisition that combines the capabilities of Large Language Models with well-established cognitive learning principles. By integrating real-time content generation, personalized difficulty

adjustment, and structured memory reinforcement techniques, the system overcomes many of the limitations found in traditional static language learning platforms.

The comparative evaluation indicates that learners using the adaptive AI framework achieved 23.6% faster learning efficiency and 84% assessment accuracy. These results demonstrate the system's effectiveness in reducing cognitive load and strengthening long-term retention.

Overall, the findings suggest that intelligent and adaptive tutoring systems can offer a meaningful improvement over conventional rote-based learning methods. By aligning AI technology with cognitive science principles, this approach provides a more efficient, engaging, and scientifically grounded model for modern language education.

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